

HOMEWORK 6

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 10/27/2023

Problem 1. Consider Runge's function, which is

$$f(x) = \frac{1}{1 + 25x^2}, \quad -1 \leq x \leq 1.$$

- (a) Implement a function to find the n -th order interpolating polynomial p_n for f (submit your code with this problem). Plot the function $f(x)$ and $p_n(x)$ on the interval $[-1, 1]$, using the uniformly spaced points $x_j = -1 + \frac{2j}{n}$, $j = 0, 1, \dots, n$, where $n = 4, 8$, and 16 .
- (b) Repeat part (a) using the Chebyshev points $x_j = \cos\left(\frac{2j+1}{2(n+1)}\pi\right)$, for $j = 0, 1, \dots, n$, where $n = 4, 8$, and 16 .
- (c) Let $w(x) = (x - x_0)(x - x_1)\dots(x - x_n)$. Plot $|w(x)|$ for $n = 4$ (i.e. $x_0, x_1, \dots, x_4$) that are:
- (i) uniform points,
 - (ii) Chebyshev points,
- on the interval $[-1, 1]$. Discuss the graphs you have obtained.

Problem 2. Given points $(x_0, y_0), \dots, (x_n, y_n) \in \mathbb{R}^2$ with $x_i \neq x_j$ for $i \neq j$, let $h_i(x) = \prod_{i \neq j} \frac{x - x_j}{x_i - x_j}$, so that the Lagrange polynomial interpolant of these points is given by

$$p(x) = \sum_{i=0}^n y_i h_i(x).$$

Prove that

(a)

$$\sum_{i=0}^n h_i(x) = 1$$

(b)

$$\sum_{i=0}^n h_i(0) x_i^k = \begin{cases} 1 & \text{for } k = 0, \\ 0 & \text{for } k = 1, 2, \dots, n, \\ (-1)^n x_0 x_1 \dots x_n & \text{for } k = n + 1, \end{cases}$$

Problem 3. Given points $(x_0, y_0), \dots, (x_n, y_n) \in \mathbb{R}^2$ with $x_i \neq x_j$ for $i \neq j$, let $h_i(x) = \prod_{i \neq j} \frac{x - x_j}{x_i - x_j}$, so that the Lagrange polynomial interpolant of these points is given by

$$p(x) = \sum_{i=0}^n y_i h_i(x).$$

Consider the approximation

$$q(x) = \sum_{i=0}^n y_i (h_i(x))^2.$$

It is not hard to see that this polynomial interpolates these points as well, since $(h_i(x_j))^2 = \begin{cases} 1^2 & i = j \\ 0^2 & i \neq j \end{cases}$

- (a) What is the degree of $q(x)$?
- (b) Suppose the points y_i are the values of a function f , (i.e., are given by $y_i = f(x_i)$ for some function $f(x)$). Show that q has the property that if f is positive for all x , then $q(x)$ is also positive for all x .
- (c) Show, by way of a simple example, that the usual interpolating polynomial does not have the property described in part (b).

Problem 4. This problem shows how to use interpolation as a means for root finding (inverse interpolation). We want to find a zero x^* of the function $f(x) = \cos(x) - x$. We know that

$$f(0.6) = 0.22534, \quad f(0.7) = 0.06484, \quad f(0.9) = -0.27839.$$

Let $g(y) = f^{-1}(y)$ be the inverse function of $f(x)$. The unknown zero x^* can then be expressed as $x^* = g(0)$.

- (a) Construct the quadratic polynomial $p_2(y)$ that interpolates g (not f !) at the given points. Determine the divided difference table using only 5 decimal digits.
- (b) Determine $p_2(0)$ and use it as an approximation for x^* .
- (c) Recall that the error bound for the interpolating polynomial $p_n(x)$ of g is given by

$$g(x) - p_n(x) = \frac{g^{(n+1)}(\zeta)}{(n+1)!} \prod_{i=0}^n (x - x_i),$$

where ζ is some point in the smallest interval containing the x_i 's and x lies in this interval as well. Give a theoretical estimate for the error $|p_2(0) - x^*|$ by using the following fact:

$$g'''(f(x)) = 3f''(x)^2 f'(x)^{-5} - f'''(x) f'(x)^{-4}.$$

- (d) Suppose that $f(0.8) = -0.10329$, using this data point together with the quadratic polynomial $p_2(y)$ to determine the polynomial $p_3(y)$. Compute $|p_2(0) - p_3(0)|$. This is an estimate for the true error.